

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Cross-Text Connections	Hard

Question

Text 1

Fossils of the hominin *Australopithecus africanus* have been found in the Sterkfontein Caves of South Africa, but assigning an age to the fossils is challenging because of the unreliability of dating methods in this context. The geology of Sterkfontein has caused soil layers from different periods to mix, impeding stratigraphic dating, and dates cannot be reliably imputed from those of nearby animal bones since the bones may have been relocated by flooding.

Text 2

Archaeologists used new cosmogenic nuclide dating techniques to reevaluate the ages of *A. africanus* fossils found in the Sterkfontein Caves. This technique involves analyzing the cosmogenic nucleotides in the breccia—the matrix of rock fragments immediately surrounding the fossils. The researchers assert that this approach avoids the potential for misdating associated with assigning ages based on Sterkfontein’s soil layers or animal bones.

Based on the texts, how would the researchers in Text 2 most likely respond to the underlined portion in Text 1?

Answer

- A. They would emphasize the fact that the *A. africanus* fossils found in the Sterkfontein Caves may have been corrupted in some way over the years.
- B. They would contend that if analyses of surrounding layers and bones in the Sterkfontein Caves were combined, then the dating of the fossils there would be more accurate.
- C. They would argue that their techniques are better suited than other methods to the unique challenges posed by the Sterkfontein Caves.
- D. They would claim that cosmogenic nuclide dating is reliable in the context of the Sterkfontein Caves because it is applied to the fossils directly.

Correct Answer: C

Rationale

Choice C is the best answer. Text 2 states that the researchers used cosmogenic nuclide dating to "avoid the potential for misdating" caused by the geology of Sterkfontein, which Text 1 describes as "challenging" and unreliable.

Choice A is incorrect. Neither text suggests that the *A. africanus* fossils have been "corrupted," but only that traditional dating methods are difficult in Sterkfontein because of floods and soil mixing where the fossils were found. Nothing is implied to have compromised the fossils themselves. Choice B is incorrect. This choice misreads Text 2. Text 2 agrees that stratigraphy and other methods are prone to error in the context of Sterkfontein: there’s a "potential for misdating" when evaluating age based on soil layers and bones. Choice D is incorrect. Text 2 does not state that cosmogenic nuclide dating is applied to the fossils directly but rather to the breccia that surrounds them.